Clauses And Operators

Create Database:

```
CREATE DATABASE University;
USE University;
```

Create Table:

```
CREATE TABLE Students (student_id INT PRIMARY KEY, first_name VARCHAR(50) NOT NULL,
last_name VARCHAR(50) NOT NULL, department VARCHAR(100) NOT NULL, email VARCHAR(100), gpa DECIMAL(3,2));
DESCRIBE Students;
```

Field	Type	Null	Key	Default
student_id	int	NO	PRI	NULL
first_name	varchar(50)	NO		NULL
last_name	varchar(50)	NO		NULL
department	varchar(100)	NO		NULL
email	varchar(100)	YES		NULL
gpa	decimal(3,2)	YES		NULL

1: DISTINCT: List unique departments:

```
SELECT DISTINCT Department FROM students;
```

Department
Computer Science
Mechanical Engineering
Electrical Engineering
Civil Engineering
Information Technology
Electronics

2: IS NULL & NOT NULL: Students with NULL and NOT NULL emails:

```
SELECT * FROM Students WHERE email IS NULL;
SELECT * FROM Students WHERE email IS NOT NULL;
```

IS NULL:

student_id	first_name	last_name	department	email	gpa
2	Priya	Sharma	Mechanical Engineering	NULL	3.20
5	Vikram	Rao	Computer Science	NULL	3.10
9	Suresh	Pillai	Mechanical Engineering	NULL	2.70
12	Lakshmi	Krishnan	Information Technology	NULL	3.40
16	Pooja	Mishra	Civil Engineering	NULL	2.85
19	Deepak	Bose	Electronics	NULL	2.60

IS NOT NULL:

student_id	first_name	last_name	department	email	gpa
1	Arun	Kumar	Computer Science	arun.kumar@univ.edu	3.45
3	Ravi	Menon	Electrical Engineering	ravi.menon@univ.edu	2.95
4	Sneha	Patel	Civil Engineering	sneha.patel@univ.edu	3.60
6	Meera	Nair	Information Technology	meera.nair@univ.edu	3.75
7	Karthik	Iyer	Electronics	karthik.iyer@univ.edu	2.80
8	Divya	Joshi	Computer Science	divya.joshi@univ.edu	3.90
10	Anita	Reddy	Civil Engineering	anita.reddy@univ.edu	3.25
11	Rahul	Verma	Computer Science	rahul.verma@univ.edu	3.55
13	Ajay	Singh	Electronics	ajay.singh@univ.edu	2.95
14	Neha	Kapoor	Computer Science	neha.kapoor@univ.edu	3.80
15	Manoj	Das	Mechanical Engineering	manoj.das@univ.edu	3.05
17	Santosh	Gupta	Computer Science	santosh.gupta@univ.edu	3.50

3: IN, BETWEEN, NOT BETWEEN

```
SELECT * FROM Students WHERE GPA BETWEEN 2 AND 3;
SELECT * FROM Students WHERE GPA NOT BETWEEN 2 AND 3;
```

BETWEEN:

student_id	first_name	last_name	department	email	gpa
3	Ravi	Menon	Electrical Engineering	ravi.menon@univ.edu	2.95
7	Karthik	Iyer	Electronics	karthik.iyer@univ.edu	2.80
9	Suresh	Pillai	Mechanical Engineering	NULL	2.70
13	Ajay	Singh	Electronics	ajay.singh@univ.edu	2.95
16	Pooja	Mishra	Civil Engineering	NULL	2.85
19	Deepak	Bose	Electronics	NULL	2.60

NOT BETWEEN:

student_id	first_name	last_name	department	email	gpa
1	Arun	Kumar	Computer Science	arun.kumar@univ.edu	3.45
2	Priya	Sharma	Mechanical Engineering	NULL	3.20
4	Sneha	Patel	Civil Engineering	sneha.patel@univ.edu	3.60
5	Vikram	Rao	Computer Science	NULL	3.10
6	Meera	Nair	Information Technology	meera.nair@univ.edu	3.75
8	Divya	Joshi	Computer Science	divya.joshi@univ.edu	3.90
10	Anita	Reddy	Civil Engineering	anita.reddy@univ.edu	3.25
11	Rahul	Verma	Computer Science	rahul.verma@univ.edu	3.55
12	Lakshmi	Krishnan	Information Technology	NULL	3.40
14	Neha	Kapoor	Computer Science	neha.kapoor@univ.edu	3.80
15	Manoj	Das	Mechanical Engineering	manoj.das@univ.edu	3.05